

1. Introduction and Literature Survey

1.1 Introduction

Floods are one of the most frequent and devastating natural disasters, causing significant loss of life, damage to infrastructure, and disruption of livelihoods, especially in rural and low-lying areas. Traditional flood monitoring systems mainly rely on centralized data and regional forecasts, which often lack real-time, location-specific accuracy. This limitation reduces the effectiveness of early warning systems and delays preventive actions.

With the advancement of the Internet of Things (IoT), it is now possible to develop smart, low-cost, and real-time monitoring solutions. This project presents an IoT-based Flood Monitoring and Cause Classification System designed to provide hyper-local flood detection and short-term prediction. The system uses multiple sensors such as water level, rainfall, soil moisture, and flow rate sensors to continuously monitor environmental conditions.

The collected data is processed using an ESP32 microcontroller to calculate flood risk and identify possible causes such as heavy rainfall, river overflow, drainage blockage, or dam release. Additionally, a prediction model estimates flood conditions 20–40 minutes in advance, enabling early warnings. Alerts are sent via GSM communication, and real-time data is uploaded to the cloud for remote monitoring.

This system aims to provide a reliable, cost-effective, and scalable solution for early flood detection, helping to minimize damage and improve disaster preparedness in vulnerable regions.

1.2 Problem Statement

Floods pose a major threat to life and infrastructure, particularly in rural and low-lying areas where monitoring systems are limited or unavailable. Existing flood forecasting systems mainly provide regional-level predictions and often lack real-time, locationspecific data. This results in delayed warnings and insufficient time for preventive actions.

Moreover, current systems do not effectively identify the exact cause of flooding, such as heavy rainfall, river overflow, drainage blockage, or dam release. The absence of short-term prediction and reliable alert mechanisms further increases the risk of damage and loss.

Therefore, there is a need for a cost-effective, real-time, and IoT-based flood monitoring system that can provide hyper-local data, accurately assess flood risk, classify causes, and generate timely alerts to improve disaster preparedness and response.

1.3 Objective:

1. To measure real-time water level using an ultrasonic sensor.
2. To monitor rainfall intensity using a rain sensor.
3. To analyze soil saturation using a capacitive soil moisture sensor.
4. To measure water velocity using a flow sensor.
5. To calculate flood risk using multi-parameter weighted scoring.
6. To classify flood causes (Heavy Rain, River Overflow, Drain Blockage, Dam Release).
7. To implement quadratic trend extrapolation for 20–40-minute prediction.
8. To send SMS alerts using a GSM module.
9. To upload real-time data to Firebase cloud.
10. To reduce false positives using hysteresis logic.

1.4 Scope of Project

The proposed IoT-based flood monitoring system is designed for hyper-local deployment in flood-prone rural and semi-urban areas where centralized monitoring infrastructure is limited or unavailable. The system focuses on short-term flood prediction within a 20–40-minute window, enabling timely early warnings to reduce damage to life and property. It can be installed near riverbanks, low-lying roads, bridges, drainage channels, and agricultural lands to monitor critical environmental parameters in real time.

The system operates as a standalone unit using GSM communication, ensuring functionality even in areas without continuous internet connectivity. Its modular and scalable architecture allows multiple nodes to be deployed across different locations to form a distributed flood monitoring network. Although the system provides effective short-term forecasting and realtime alerts, it does not support long-term hydrological prediction. Performance may depend on GSM network availability, and sensors require periodic calibration to maintain measurement accuracy.

1.5 Literature Survey

Flood monitoring and prediction have been widely studied using various technologies, including traditional hydrological models and modern IoT-based systems. Several researchers have proposed solutions to improve early warning systems and reduce flood-related damage.

Myint M. M. and Thein T. L. L. (2025) developed a real-time flood monitoring system using IoT for urban areas. Their system focused on continuous data collection and real-time alerts, but it was mainly designed for city environments and lacked adaptability for rural deployment.

Aranganathan A. et al. (2024) proposed integrating IoT technologies for enhanced flood monitoring. Their work highlighted the importance of multiple sensor data and cloud connectivity. However, the system relied heavily on internet availability, which limits its usability in remote areas.

Zulkapli Z. B. et al. (2025) designed an IoT-based flash flood monitoring system with mobile notification alerts. While their approach improved communication efficiency, it primarily focused on alert generation and did not include detailed cause classification or prediction models.

Canillo L. J. L. and Hernandez A. A. (2021) introduced a flood risk visualization and prediction system. Their model used data analysis techniques to estimate flood risk but lacked real-time sensor integration, making it less effective for immediate response scenarios.

Zheng W. (2012) proposed a flood monitoring framework using multi-source satellite data. Although effective for large-scale monitoring, satellite-based systems are expensive and not suitable for localized, real-time applications.

Rani D. S. et al. (2020) developed a low-cost IoT-based flood monitoring system using machine learning techniques. Their work demonstrated improved prediction accuracy but increased system complexity and cost, making it less practical for low-resource areas.

From the above studies, it is observed that existing systems either focus on large-scale monitoring, require high-cost infrastructure, depend heavily on internet connectivity, or lack real-time prediction and cause classification. Hence, there is a need for a cost-effective, standalone, and real-time IoT-based system that can provide short-term prediction and accurate flood cause identification.

1.6 Methodology

The proposed IoT-based flood monitoring system works through a sequence of data collection, processing, analysis, prediction, and alert generation. The system uses multiple sensors connected to an ESP32 microcontroller to continuously monitor environmental conditions and assess flood risk in real time.

Step 1: Data Acquisition

- Water level is measured using an ultrasonic sensor.
- Rainfall intensity is detected using a rain sensor.
- Soil moisture is monitored using a capacitive soil moisture sensor.
- Water flow rate is calculated using a flow sensor.

Step 2: Data Processing

- Sensor data is collected by the ESP32.
- Noise filtering and normalization are applied to improve accuracy.

Step 3: Flood Risk Calculation

- Each parameter is assigned a specific weight.
- A weighted scoring algorithm is used to compute flood risk (0–100%).

Step 4: Cause Classification

- Rule-based logic is applied to identify the cause of flooding:
 - Heavy Rain
 - River Overflow
 - Drain Blockage
 - Dam Release

Step 5: Short-Term Prediction

- The system calculates rise rate and acceleration of water level.
- A quadratic trend model is used to predict flood conditions for the next 20–40 minutes.

Step 6: Alert Generation

- If flood risk exceeds a threshold, alerts are triggered.
- SMS notifications are sent using a GSM module.
- Alerts may also include buzzer or LED indications.

Step 7: Hysteresis Logic Implementation

- Alerts are confirmed over multiple cycles before triggering.
- This reduces false alarms caused by temporary fluctuations.

Step 8: Cloud Monitoring

- Data is uploaded to Firebase cloud for remote monitoring.
- Users can track real-time conditions and prediction history.

2. System Specification and Block Schematic

2.1 Overall System Description

The proposed system is an IoT-based Flood Monitoring and Cause Classification System designed to monitor environmental conditions in real time and provide early flood warnings. It uses multiple sensors to collect data such as water level, rainfall, soil moisture, and flow rate. An ESP32 processes this data to calculate flood risk, predict future conditions, and identify possible flood causes. Alerts are sent via GSM, and data is uploaded to the cloud for remote monitoring.

2.2 System Block Diagram

The system consists of three main sections:

- Input

Ultrasonic Sensor, Rain Sensor, Soil Moisture Sensor, Flow Sensor

- Processing

ESP32 Microcontroller, Risk Calculation Unit, Prediction Algorithm, Cause Classification Logic

- Output

GSM Module (SMS/Call Alerts), Firebase Cloud, Buzzer/LED (optional)

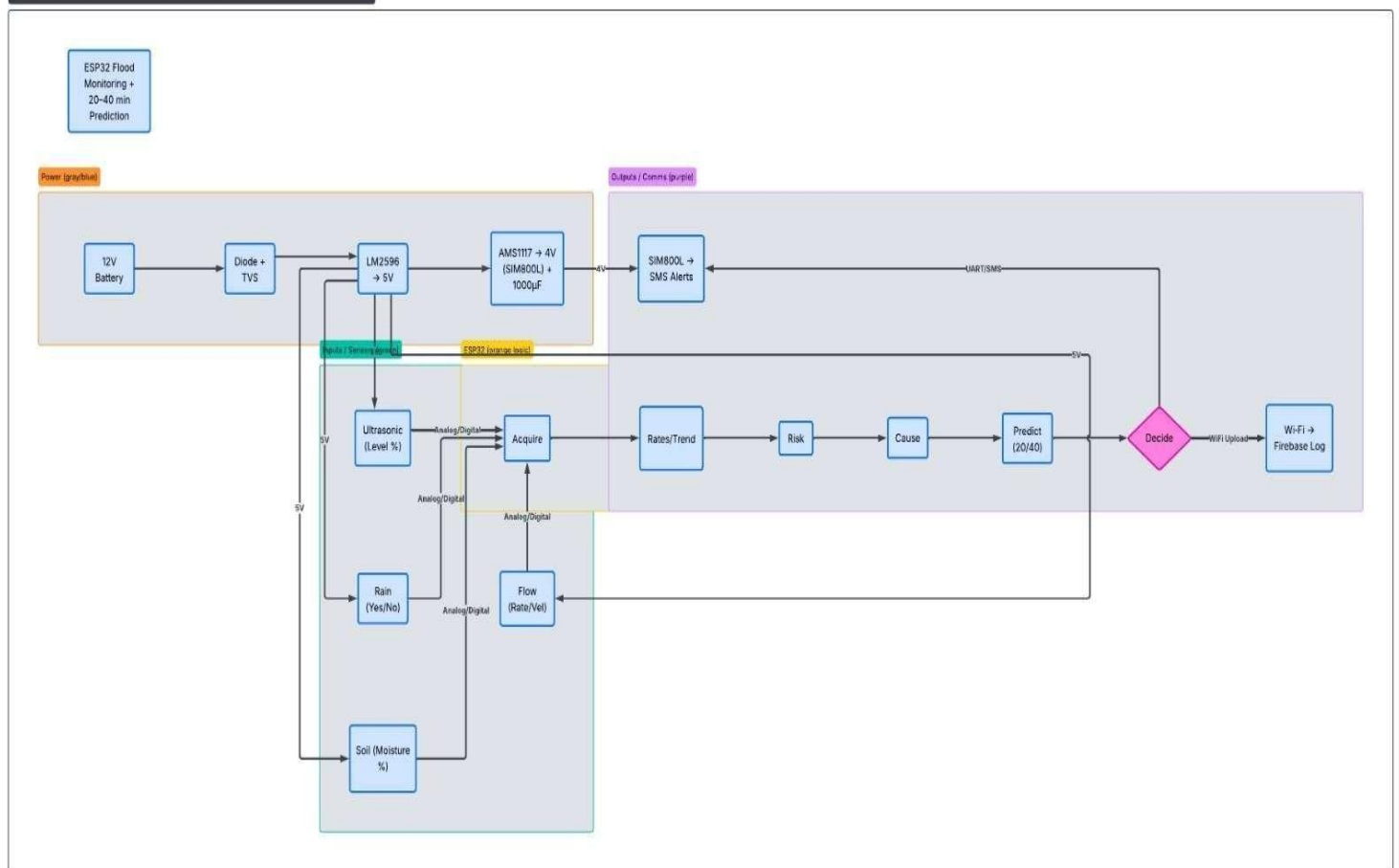


Figure No. 1 Block diagram of system

2.3 System Requirements

2.3.1 Software Requirements

- Arduino IDE
- Embedded C/C++
- Firebase Real-time Database
- Required Libraries:
 - WiFi Library
 - HTTP Client Library
 - Firebase ESP Client Library

2.3.2 Hardware Requirements

- ESP32 Microcontroller
- Ultrasonic Sensor
- Rain Sensor
- Soil Moisture Sensor

- Flow Sensor
- GSM Module
- Power Supply
- Connecting wires, PCB
- Optional: Buzzer/LED

2.4 Working Principle

The system continuously collects environmental data through sensors. The ESP32 processes this data and calculates flood risk using a weighted scoring method. Based on sensor values, the system identifies the cause of flooding using rule-based logic. It also predicts future flood conditions using a quadratic trend model.

If the flood risk exceeds a threshold or prediction indicates danger, alerts are sent via GSM. Data is also uploaded to Firebase for real-time monitoring. Hysteresis logic ensures alerts are generated only when conditions persist, reducing false alarms.

2.5 Advantages

1. Real-time flood monitoring
2. Low-cost and scalable solution
3. Works in rural areas with GSM support
4. Provides short-term prediction (20–40 minutes)
5. Reduces false alarms using hysteresis logic
6. Cloud-based monitoring

2.6 Limitations

- Depends on GSM network availability
- Requires periodic sensor calibration
- Limited to short-term prediction
- Accuracy may vary with environmental conditions

2.7 Applications

- Flood-prone rural and urban areas
- Riverbanks and dams
- Drainage monitoring systems

3. Hardware Design & Software Design

3.1 Hardware Design

The hardware design of the system is centered around the ESP32, which acts as the main processing unit. Various sensors are interfaced with the ESP32 to collect environmental data:

- Ultrasonic Sensor: Measures water level
- Rain Sensor: Detects rainfall intensity
- Soil Moisture Sensor: Measures ground saturation
- Flow Sensor: Calculates water velocity

A GSM module is used for sending SMS alerts and making emergency calls. All components are connected using appropriate power supply and interfacing circuits. The system is designed to operate continuously in outdoor conditions.

3.2 Software Design

The software is developed using Embedded C/C++ in the Arduino IDE. It controls sensor data acquisition, processing, and communication.

Main software modules include:

- Sensor data reading and calibration
- Flood risk calculation (weighted algorithm)
- Cause classification logic
- Prediction algorithm (quadratic model)
- GSM communication for alerts
- Cloud integration using Firebase

The software ensures real-time operation and efficient handling of data.

3.3 Flowchart / Algorithm

Algorithm Steps:

1. Start the system
2. Initialize sensors and GSM module
3. Read sensor data (water level, rain, soil, flow)
4. Process and normalize data

5. Calculate flood risk using weighted formula
6. Identify flood cause using rule-based logic
7. Apply prediction model (20–40 min ahead)
8. Check threshold conditions o If risk high → Send alert o Else → Continue monitoring
9. Upload data to Firebase
10. Apply hysteresis logic to avoid false alerts
11. Repeat loop continuously

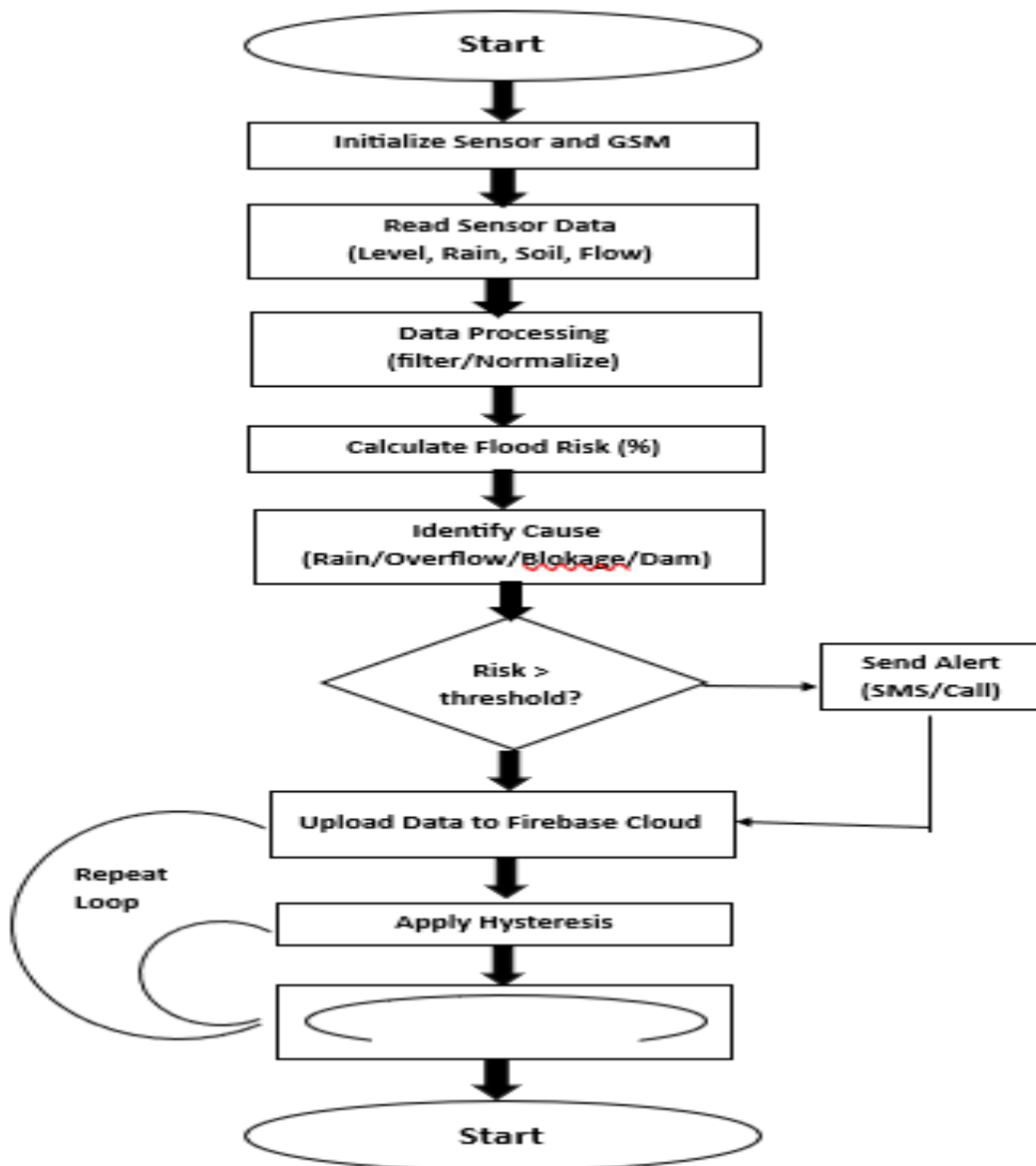


Figure No. 2 Flow Chart of system

3.4 Circuit Diagram / Architecture

The circuit consists of sensors connected to the ESP32 through GPIO pins:

- Ultrasonic sensor → Digital pins (Trigger & Echo)
- Rain sensor → Digital input
- Soil moisture sensor → Analog input
- Flow sensor → Interrupt pin
- GSM module → UART communication (TX/RX pins)

The ESP32 processes all inputs and controls output devices. Power supply is provided to all components with proper grounding.

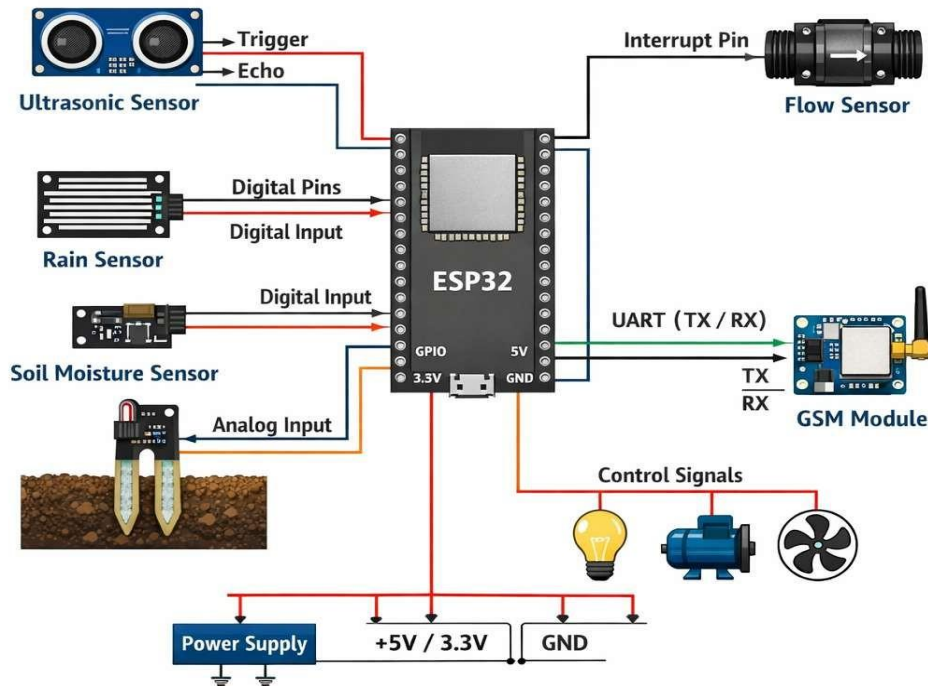


Figure No. 3 Circuit diagram of system

3.5 Implementation Details

The system is implemented by integrating both hardware and software components:

- Sensors are mounted at appropriate locations (riverbanks, drainage areas, etc.)
- ESP32 is programmed and tested using Arduino IDE
- GSM module is configured for SMS and call alerts

- Firebase is set up for real-time cloud data storage
- System is tested under different environmental conditions

The implementation ensures accurate data collection and reliable alert generation.

4. Results, Discussion and Analysis

4.1 Experimental Setup

The experimental setup consists of an IoT-based system built using an ESP32 interfaced with multiple sensors including an ultrasonic sensor, rain sensor, soil moisture sensor, and flow sensor. The system is powered using a regulated power supply, and a GSM module is used for alert communication.

Sensors are placed in a controlled environment to simulate flood conditions such as increasing water levels, rainfall presence, and varying soil moisture. The ESP32 collects real-time data, processes it, and uploads it to Firebase for monitoring.

4.2 Results (Comparison Analysis)

The system was tested under different environmental conditions, and results were observed:

- When water level and rainfall increased simultaneously → High flood risk detected
- When only water flow increased → River overflow condition identified
- When rainfall present but low flow → Drain blockage detected
- Prediction system successfully estimated flood conditions 20–40 minutes in advance

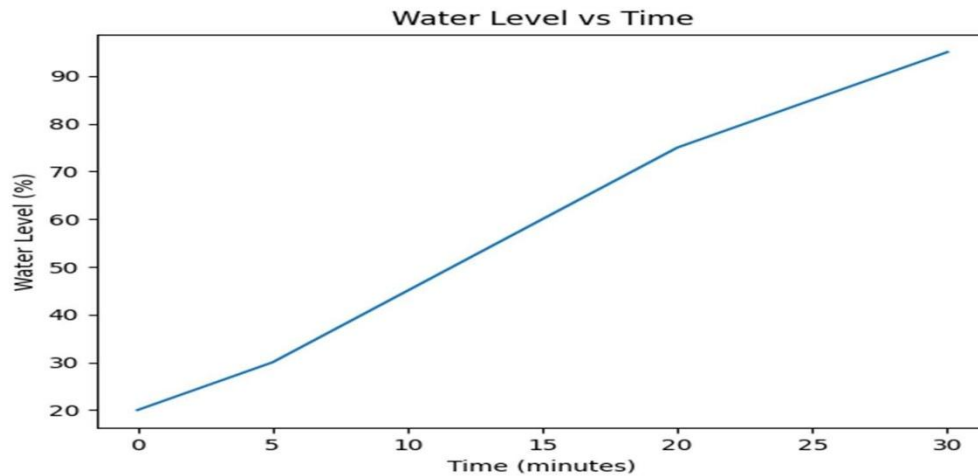
Compared to traditional systems, this model provides faster response and localized prediction, improving early warning capability.

4.3 Results Tables and Graphs

The collected data can be represented using tables and graphs such as:

- Water Level vs Time
- Flood Risk (%) vs Time
- Soil Moisture vs Rainfall
- Prediction Score vs Time

These graphs help in visualizing system performance and understanding trends in



flood conditions.

Figure No. 4 Graph analysis of result

Table No.1 Comparison of systems

Feature	Existing System	Proposed System
Monitoring Type	Regional	Local (IoT-based)
Real-Time Data	Limited	Yes
Prediction	Long-term	Short-term (20–40 min)
Cost	High	Low
Internet Dependency	High	Low (GSM based)
Cause Classification	Not available	Available

4.4 Performance Analysis

The system performance is evaluated based on:

- Accuracy: Depends on sensor calibration
- Response Time: Real-time monitoring with quick alerts
- Reliability: Improved using hysteresis logic
- Prediction Efficiency: Provides early warning within 20–40 minutes

The system performs efficiently in detecting and predicting flood conditions with minimal delay.

4.5 Comparison with Existing System

The proposed system overcomes limitations of existing systems by providing real-time, low-cost, and accurate flood monitoring

4.6 System Advantages and Disadvantages

Advantages

- Real-time monitoring
- Early flood prediction
- Low-cost and scalable
- Works in rural areas
- GSM-based alerts
- Reduced false alarms

Disadvantages

- Depends on sensor accuracy
- Requires maintenance and calibration
- Limited to short-term prediction
- GSM network dependency

5. Conclusion and Future Scope

5.1 Conclusion

The proposed IoT-based Flood Monitoring and Cause Classification System successfully provides real-time monitoring and early flood prediction using multiple environmental parameters. By integrating sensors with an ESP32, the system efficiently calculates flood risk and identifies possible causes such as heavy rainfall, river overflow, drainage blockage, and dam release.

The implementation of a short-term prediction model (20–40 minutes) helps in generating timely alerts, thereby reducing potential damage to life and property. The use of GSM communication ensures alert delivery even in areas with limited internet connectivity, while cloud integration enables remote monitoring. Overall, the system is cost-effective, reliable, and suitable for deployment in flood-prone regions.

5.2 Limitations

- The system depends on the accuracy and calibration of sensors
- GSM network availability affects alert delivery
- Limited to short-term prediction only (20–40 minutes)
- Environmental factors may affect sensor readings
- Requires regular maintenance and monitoring

5.3 Future Scope

- Integration of machine learning for improved prediction accuracy
- Addition of GPS for location-based alerts
- Mobile application for real-time monitoring
- Solar-powered system for energy efficiency
- Expansion into a large-scale distributed monitoring network
- Integration with government disaster management systems

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Appendix

1. Source Code

The system is implemented using Embedded C/C++ on the ESP32. The code includes sensor interfacing, flood risk calculation, prediction algorithm, GSM communication, and cloud data upload.

The following program is written in Embedded C/C++ using Arduino IDE for ESP32. It performs:

- Sensor data acquisition
- Flood risk calculation
- Cause classification
- Prediction logic
- GSM alert system
- Firebase cloud upload

Program

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Firebase_ESP_Client.h>

// ----- Pin Definitions ----- #define
TRIG 5
#define ECHO 18
#define RAIN 19
#define FLOW 27
#define SOIL 34

#define RXD2 16
#define TXD2 17
```

```

// ----- Global Variables ----- float
previousLevel = 0;
volatile int pulseCount = 0;

HardwareSerial gsm(2);

// ----- Interrupt Function -----
void IRAM_ATTR pulseCounter() {
    pulseCount++;
}

// ----- Setup Function ----- void
setup() {
    Serial.begin(115200);

    pinMode(TRIG,          OUTPUT);
    pinMode(ECHO, INPUT);  pinMode(RAIN,
    INPUT);  pinMode(FLOW, INPUT);
    pinMode(SOIL, INPUT);

    attachInterrupt(digitalPinToInterrupt(FLOW), pulseCounter, FALLING);

    gsm.begin(9600, SERIAL_8N1, RXD2, TXD2);

    Serial.println("System Initialized...");
}

// ----- Function: Read Water Level -----
float readWaterLevel() {  digitalWrite(TRIG,
    LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG, HIGH);

```

```

delayMicroseconds(10);
digitalWrite(TRIG, LOW);

long duration = pulseIn(ECHO, HIGH);
float distance = duration * 0.034 / 2;

float level = 100 - distance; return
constrain(level, 0, 100);
}

// ----- Main Loop ----- void
loop() {

    float waterLevel = readWaterLevel();    bool
    isRaining = digitalRead(RAIN) == LOW;    int
    soilValue = analogRead(SOIL);

    float flowRate = pulseCount / 7.5; pulseCount
    = 0;

    float soilPercent = map(soilValue, 4095, 0, 0, 100);

    // ----- Flood Risk Calculation ----- float
    floodRisk = (0.4 * waterLevel) +
                (0.3 * flowRate) +
                (0.2 * (isRaining ? 100 : 0)) +
                (0.1 * soilPercent);

    // ----- Cause Classification -----
    String cause = "Normal";

    if (isRaining && soilPercent > 70)
    cause = "Heavy Rain Flood"; else if

```

```

(flowRate > 10)    cause = "River
Overflow";  else if (isRaining &&
flowRate < 5)
    cause = "Drain Blockage";

// ----- Display Output -----  Serial.println("---
-- DATA -----");
Serial.print("Water Level: "); Serial.println(waterLevel);
Serial.print("Soil Moisture: "); Serial.println(soilPercent);
Serial.print("Flow Rate: "); Serial.println(flowRate);
Serial.print("Flood Risk: "); Serial.println(floodRisk);
Serial.print("Cause: "); Serial.println(cause);

// ----- Alert Condition -----  if
(floodRisk      >      70)      {
gsm.println("AT+CMGF=1");
    delay(1000);

    gsm.println("AT+CMGS=\"+91XXXXXXXXXXXX\"");
    delay(1000);

    gsm.println("Flood    Alert!    High    Risk    Detected.");
gsm.write(26); // CTRL+Z
}

delay(5000);
}

```

2. Pin Configuration Table

Table No. 2 Pin configuration

Component	ESP32 Pin
Ultrasonic TRIG	GPIO 5
Ultrasonic ECHO	GPIO 18
Rain Sensor	GPIO 19
Soil Moisture Sensor	GPIO 34
Flow Sensor	GPIO 27
GSM RX	GPIO 16
GSM TX	GPIO 17

3. Components List

Table No. 3 Components List

Sr. No	Component Name	Quantity
1	ESP32 Microcontroller	1
2	Ultrasonic Sensor	1
3	Rain Sensor	1
4	Soil Moisture Sensor	1
5	Flow Sensor	1
6	GSM Module	1
7	Power Supply	1
8	Connecting Wires	As required

4. Sample Output (Serial Monitor)

Water Level: 75%

Rain Detected: YES

Soil Moisture: 68%

Flow Rate: 12 L/min

Flood Risk: 82%

Cause: Heavy Rain Flood

Prediction: High Probability Flood in 20–40 min

5. Abbreviations

- IoT – Internet of Things
- GSM – Global System for Mobile Communication
- GPIO – General Purpose Input Output
- SMS – Short Message Service

6. Additional Notes

- Ensure proper grounding of all components
- Calibrate sensors before deployment
- Use external power supply for GSM module
- Maintain system regularly for accurate results

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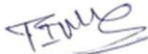
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